AGNEL CHARITIES

FR. C. RODRIGUES INSTITUTE OF TECHNOLOGY

Agnel Technical Education Complex Sector 9-A, Vashi,

(An Autonomous Institute & Permanently Affiliated to University of Mumbai)

CCL Mini Project

SUBJECT	Cloud Computing Lab	
Branch & Semester	COMP-VI	
	Roll Number	Name of the Student
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Subject Incharge Signature with Date		

INTRODUCTION

Brainline is an innovative mobile application designed to raise awareness about brain strokes and help individuals assess their risk of experiencing one. With a focus on user engagement and education, Brainline features a unique Riskometer, which calculates a personalized stroke risk score based on various health factors. The app offers an enhanced user interface (UI), ensuring that users have a smooth and intuitive experience while navigating through the app's features. By combining education, prevention tips, and real-time risk assessments, Brainline empowers users to take control of their health and make informed decisions regarding stroke prevention. This app aims to promote early detection, better understanding, and proactive measures against brain strokes.

CLOUD SERVICE PROVIDED BY APPLICATION

Features :

The frontend of the application is developed using Bootstrap, a widely-used CSS framework that ensures a responsive and visually appealing design. The key features of the frontend are as follows:

1. **Responsive Design:**
- Bootstrap's grid system and pre-built components enable the app to be responsive across all screen sizes, from mobile phones to desktops. This ensures the application adapts seamlessly to different devices.
2. **User Interface (UI):**
- The UI is designed with Bootstrap's ready-to-use components like buttons, navigation bars, forms, and cards. These components help in quickly creating a clean and consistent interface with minimal custom styling.
3. **State Management:**
- For managing the global state of the app, React Context API or Redux is used, ensuring smooth data flow and interaction between different parts of the application.

Hosting :

The project is hosted on an Google Cloud Platform VM Instance, which provides scalable compute resources to handle the application's business logic, database interactions, and user requests. Key features of the VM Instance include:

1. **Customizable Machine Types:**

- GCP VM instances allow you to select from a wide range of predefined machine types or create custom machine types tailored to your specific needs. You can allocate the desired number of vCPUs and amount of RAM, ensuring optimal performance while managing costs effectively.

2. **High Scalability and Flexibility**

- Compute Engine supports manual and automatic scaling, making it suitable for both small applications and large-scale deployments. You can create managed instance groups with autoscaling and load balancing to distribute traffic efficiently and handle increased user demand smoothly.

3. **Persistent Storage with Snapshots:**

- VM instances use Persistent Disks that provide durable and high-performance storage. These disks can be easily resized and backed up using snapshots, ensuring data safety and supporting disaster recovery strategies. Disks can also be detached and reused across different instances.

4. **Robust Security and Access Control:**

- Google Cloud ensures enterprise-grade security for VM instances. Features include firewall rules, SSH key management, Identity and Access Management (IAM) for role-based access, and Shielded VMs that provide protection against rootkits and boot-level attacks. Data is encrypted at rest and in transit by default.

5. **Global Deployment and High Availability:**

- GCP allows you to deploy VM instances across multiple regions and zones worldwide, enhancing availability and reducing latency. This geographical distribution also supports fault tolerance and disaster recovery, ensuring minimal downtime and improved reliability for global users.